Interference and the price of a feature: what replaces Conjecture 20

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Abstract

Conjecture 20 (the unrestricted maximum $L_{2,b}$-margin of product networks equals the interference-free value $\nu^{1-s}B_p\Gamma_s(p)$) is false: at $p=3$, $b=2$ certified networks beat it for $n=11,13,14$, and for $n=12$ on $b\in[2,2.13]$. This note places the counterexamples inside the paper's framework. (i) We write every margin condition in terms of one complex *feature field* $\Phi_l:\mathbb{Z}_p^n\to\mathbb{C}$ per frequency, and show that a single feature has margin $\geq c$ exactly when its field stays in the cone $c+\{|\arg z|\leq\pi/p\}$. (ii) We define the *price of a feature* $\pi_{n,b}(p)\in[\nu^{1-s},1]$, the best normalised margin of a single feature with interference allowed, and prove the *price–programme theorem* $\gamma_b^*\geq\pi_{n,b}(p)B_p\Gamma_s(p)$, realised by networks whose amplitudes follow a maximiser $y^\star$ of the programme (so they use exactly $K^*(s,p)$ features). Theorem 8 of the paper is the case $\pi=\nu^{1-s}$; at $p=3$ the inequality is an equality. Hence the counterexamples change the *price of a feature*, not the spectral allocation. (iii) We identify what the counterexamples exploit and prove it is unavailable to single-frequency networks at $p\geq5$: there the feature field is *circular* ($\mathbb{E}(\Phi-\mathbb{E}\Phi)^2=0$) and the cone has half-angle $\pi/p<\pi/4$, which forces a margin loss $\geq k_p\sigma$ proportional to the interference σ. At $p=3$ the field need not be circular because $2\equiv-1\pmod 3$, and the certified $n=11$ network uses exactly this freedom: interference 258 times the feature's energy, with imaginary fluctuations carrying 1.78 times the energy of the real ones, parked on the 60° cone edges. Whether the loss always outweighs the saving at $p\geq5$ reduces to one constant $k^*(n,s)$ (estimated numerically). (iv) We list what in the paper is untouched, and give the corrected statement of Appendix D.

1 Setting

p is an odd prime, $\rho=e^{2\pi i/p}$, $F=\{1,\dots,(p-1)/2\}$, $\nu=n+1$, $2\leq b\leq\nu$, $s=\nu/b$, $r=1/s$, $\kappa_l(\delta)=1-\cos(2\pi l\delta/p)$ and $\kappa_{\min}(p)=1-\cos(2\pi/p)=\min_{\delta\neq0}\kappa_l(\delta)$. A product network is $f(x)[c]=\sum_i w_i(c)\prod_{t=1}^n u_{it}(x_t)$, its margin is $h=\min_x\min_{c\neq y(x)}(f(x)[y(x)]-f(x)[c])$, $y(x)=\sum_t x_t$, and $\gamma=h/\|\theta\|_{2,b}^\nu$; γ_b^* is the supremum over all widths. $B_p=2(2p\nu)^{-\nu/2}$ and $C_b^{\nu/b}=B_p$ is the single-neuron budget (paper, Appendix C): $\sum_j|\beta_{ij}|^r\leq C_b\|\omega_i\|^b$. β_l, $\beta_{l,T}$ are the paper's clean and sign-flipped coefficients (Section 4), and $\beta_{l,T}=\sum_i\beta_{il}\xi_{il}^T$ with $|\xi_{ilt}|=1$ (paper, Lemma 6). The interference-free (IF) value is $\gamma_{b,\mathrm{IF}}^*=\nu^{1-s}B_p\Gamma_s(p)$ (Theorem 8).

2 Feature fields and the cone

Definition 1. For a network θ and $l\in F$ let $\Phi_l(x):=\frac{2}{p}\,\rho^{l\,y(x)}\sum_c f(x)[c]\rho^{-lc}$.

Lemma 2 (decomposition)**.** $f(x)[c] = \bar{f}(x) + \sum_{l\in F} \operatorname{Re}\big(\Phi_l(x)\rho^{l(c-y(x))}\big)$*, hence for every* x *and* $\delta \neq 0$

$$g(x,\delta) := f(x)[y(x)] - f(x)[y(x)-\delta] = \sum_{l\in F} \operatorname{Re}\big(\Phi_l(x)(1-\rho^{-l\delta})\big).$$

Moreover $\mathbb{E}_x \Phi_l = \overline{\beta_l}$*, and for a product network* $\Phi_l(x) = \frac{2}{p}\sum_i \hat{w}_i(l)\prod_t u_{it}(x_t)\rho^{lx_t}$*.*

Proof. Fourier inversion of the real function $c \mapsto f(x)[c]$ gives $f(x)[c] = \bar{f}(x) + \sum_{l\in F}\frac{2}{p}\operatorname{Re}(\hat{f}_x(l)\rho^{lc})$ with $\hat{f}_x(l) = \sum_c f(x)[c]\rho^{-lc}$, and $\frac{2}{p}\hat{f}_x(l)\rho^{lc} = \Phi_l(x)\rho^{l(c-y)}$. For the mean,

$$\beta_l = 2p^{-\nu}\sum_{x,c} f(x)[c]\rho^{-l\sum_t x_t + lc} = 2p^{-\nu}\sum_x \overline{\rho^{ly(x)}\hat{f}_x(l)} = \overline{\mathbb{E}_x \Phi_l},$$

using $\rho^{l\sum_t x_t} = \rho^{ly(x)}$. The last formula is $\rho^{ly(x)} = \prod_t \rho^{lx_t}$. □

Lemma 3 (cone)**.** *Let* $z \in \mathbb{C}$*,* $c \in \mathbb{R}$*,* $l \in F$*, and* $K_p = \{w : |\operatorname{Im} w| \le \tan(\pi/p)\operatorname{Re} w\}$*. Then* $\operatorname{Re}(z(1-\rho^{-l\delta})) \ge c\,\kappa_l(\delta)$ *for all* $\delta \neq 0$ *if and only if* $z \in c + K_p$*, i.e.* $\operatorname{cm}(z) := \operatorname{Re} z - \cot(\pi/p)|\operatorname{Im} z| \ge c$*. If* $\operatorname{cm}(z) \ge 0$ *then* $\min_{\delta\neq 0}\operatorname{Re}(z(1-\rho^{-l\delta})) = \kappa_{\min}(p)\operatorname{cm}(z)$*.*

Proof. Write $z = a + ib$ and $\vartheta = 2\pi l\delta/p$; as δ runs over $\mathbb{Z}_p\setminus\{0\}$, $\vartheta/2$ runs over $\{\pi k/p : 1 \le k \le p-1\}$. Since $1 - e^{-i\vartheta} = 2\sin(\vartheta/2)\,(\sin(\vartheta/2) + i\cos(\vartheta/2))$,

$$\operatorname{Re}(z(1-e^{-i\vartheta})) = 2\sin(\vartheta/2)\big(a\sin(\vartheta/2) - b\cos(\vartheta/2)\big), \qquad \kappa = 2\sin^2(\vartheta/2),$$

and $\sin(\vartheta/2) > 0$, so the inequality reads $a - c \ge b\cot(\vartheta/2)$. This is affine in $\cot(\vartheta/2) \in [-\cot(\pi/p), \cot(\pi/p)]$, whose extreme values occur at $k = 1, p-1$; so it holds for all k iff $a - c \ge |b|\cot(\pi/p)$. For the last claim take $c = \operatorname{cm}(z)$: every gap is $\ge c\kappa_l(\delta) \ge c\kappa_{\min}$, and the extreme $k \in \{1, p-1\}$ with $b\cot(\vartheta/2) = |b|\cot(\pi/p)$ has $\kappa = \kappa_{\min}$ and equality. □

For a single feature the margin is therefore governed by the cone $c + K_p$ of half-angle π/p: 60° at $p = 3$, 36° at $p = 5$, 25.7° at $p = 7$.

3 The price of a feature and the price–programme theorem

Definition 4. A network is *single-feature* if every output vector w_i lies in $\operatorname{span}\{\mathbf{1}, \cos(2\pi\cdot/p), \sin(2\pi\cdot/p)\}$; then only Φ_1 is non-zero. The *price of a feature* is

$$\pi_{n,b}(p) := \sup\Big\{\frac{\min_x \operatorname{cm}(\Phi_1(x))}{B_p\,\|\theta\|_{2,b}^{\nu}} :\ \theta \text{ single-feature, any width}\Big\}.$$

By Lemma 3, the largest normalised margin of a single-feature network is $\kappa_{\min}(p)\,B_p\,\pi_{n,b}(p)$.

Proposition 5. $\nu^{1-s} \le \pi_{n,b}(p) \le 1$*.*

Proof. Lower bound: the paper's block (Proposition 9) at $j = 1$ has $\Phi_1 \equiv A > 0$ and attains Theorem 8 with $y = e_1$, i.e. normalised margin $\nu^{1-s}B_p\kappa_{\min}$. Upper bound: $\operatorname{cm}(\Phi_1(x)) \le \operatorname{Re}\Phi_1(x)$, so $\min_x \operatorname{cm} \le \operatorname{Re}\mathbb{E}\Phi_1 = \operatorname{Re}\overline{\beta_1} \le \sum_i |\beta_{i1}| \le (\sum_i |\beta_{i1}|^r)^{1/r} \le (C_b\|\theta\|_{2,b}^b)^s = B_p\|\theta\|_{2,b}^{\nu}$, using $r \le 1$. □

Theorem 6 (price–programme)**.** *For every odd prime p, $n \geq 1$, $2 \leq b \leq \nu$ and every $y \in \mathbb{R}^F_{\geq 0}$ with $\sum_l y_l^{1/s} \leq 1$,*

$$\gamma_b^* \;\geq\; \pi_{n,b}(p)\, B_p \min_{\delta \neq 0} \sum_l y_l \kappa_l(\delta), \qquad \textit{in particular} \qquad \gamma_b^* \;\geq\; \pi_{n,b}(p)\, B_p\, \Gamma_s(p).$$

The networks realising the bound are frequency-relabelled copies of one single-feature network, one copy per $l \in \operatorname{spt}(y)$, scaled by y_l; at $y = y^\star$ they use exactly $K^*(s,p)$ features with amplitudes proportional to $y^\star$.

Proof. Let θ be single-feature with $c := \min_x \operatorname{cm}(\Phi_1(x)) > 0$ and $P := \|\theta\|_{2,b}^b$. For $l \in F$ define $\theta^{(l)}$ by $u_{it}^{(l)}(a) = u_{it}(la)$, $w_i^{(l)}(c') = w_i(lc')$. Since $a \mapsto la$ permutes $\mathbb{Z}_p$, $\|\theta^{(l)}\|_{2,b} = \|\theta\|_{2,b}$, and $f^{(l)}(x)[c'] = f(lx)[lc']$ with $y(lx) = l\, y(x)$, so $g^{(l)}(x,\delta) = g(lx, l\delta) \geq c\,\kappa_1(l\delta) = c\,\kappa_l(\delta)$ by Lemma 3. Scaling every weight of $\theta^{(l)}$ by $y_l^{1/\nu}$ multiplies its table by y_l and its $\|\cdot\|_{2,b}^b$ by y_l^r. The union Θ of the scaled copies has $g_\Theta(x,\delta) = \sum_l y_l g^{(l)}(x,\delta) \geq c \sum_l y_l \kappa_l(\delta)$ and $\|\Theta\|_{2,b}^b = P \sum_l y_l^r \leq P$, so $\gamma(\Theta) \geq c \min_\delta \sum_l y_l \kappa_l(\delta)/P^s$. Taking the supremum over θ gives the claim. □

Corollary 7 ($p = 3$)**.** *$F = \{1\}$, so every network is single-feature and $\gamma_b^*(3) = \frac{3}{2} B_3\, \pi_{n,b}(3)$. The certified networks give $\pi_{11,2}(3) \geq 1.0543 \cdot 12^{-5}$, $\pi_{13,2}(3) \geq 1.1091 \cdot 14^{-6}$, $\pi_{14,2}(3) \geq 1.4293 \cdot 15^{-13/2}$, and $\pi_{12,b}(3) > 13^{1-13/b}$ for $b \in [2, 2.13]$, $\pi_{11,b}(3) > 12^{1-12/b}$ for $b \in [2, 2.02]$.*

Corollary 8 (failure is a statement about the price)**.** *Conjecture 20 holds at (p,n,b) only if $\pi_{n,b}(p) = \nu^{1-s}$. Whenever a single feature can be bought below the interference-free price ($\pi_{n,b}(p) > \nu^{1-s}$), the unrestricted optimum exceeds $\gamma^*_{b,\mathrm{IF}}$ by at least the factor $\pi_{n,b}(p)/\nu^{1-s}$* with the same spectral allocation $y^\star$ and the same number $K^*(s,p)$ of features.

Remark 9 (what the counterexamples do not touch)**.** At $p = 3$ there is a single feature, so the counterexamples change only the constant in front of $B_3\Gamma_s(3)$. Theorem 4 (universal bound, i.e. $\pi \leq 1$), Theorem 8 (exact IF value and IF structure), Theorem 10 (order $p^{-\nu/2}(\ln p)^{1-s}$ of the unrestricted optimum), Theorem 11 (programme maximisers use $\Theta(\log p)$ features) and Corollary 12 (unrestricted near-maximisers need $\Omega(\log p)$ features) are statements that the counterexamples do not contradict. The smallest known counterexample has $\nu = 12$; the paper's experiments use $\nu \leq 5$ and $p \geq 11$.

4 Why $p = 3$: circular feature fields

Definition 10. A network is *single-frequency* (at frequency 1) if every u_{it} and every w_i lies in $\operatorname{span}\{\mathbf{1}, \cos(2\pi \cdot /p), \sin(2\pi \cdot /p)\}$: constant components are allowed. At $p = 3$ every network is single-frequency. Relabelling $a \mapsto la$ (as in the proof of Theorem 6) transfers everything below to a network using any single frequency l.

Lemma 11 (support)**.** *For a single-frequency network, $\Phi_1(x) = \sum_{e \in \{0,1,2\}^n} a_e \rho^{e \cdot x}$ with $a_0 = \overline{\beta_1}$ and $a_{2\cdot \mathbf{1}_T} = \overline{\beta_{1,T}}$ for every $T \subseteq [n]$; the coefficients with some $e_t = 1$ come from constant input components.*

Proof. $u(a)\rho^a = p^{-1}\big(\hat u(-1) + \hat u(0)\rho^a + \hat u(1)\rho^{2a}\big)$; expand the product in Lemma 2. The term taking $\hat u_{it}(1)$ for $t \in T$ and $\hat u_{it}(-1)$ otherwise equals $\frac{2}{p} p^{-n} \sum_i \hat w_i(1) \prod_{t \in T} \hat u_{it}(1) \prod_{t \notin T} \hat u_{it}(-1) = \overline{2p^{-\nu} \hat f_\theta(k; -1)} = \overline{\beta_{1,T}}$ with $k_t = -1$ on T and $+1$ off T. □

Lemma 12 (circularity)**.** *If $p \geq 5$, every single-frequency network satisfies $\mathbb{E}_x(\Phi_1 - \mathbb{E}\Phi_1)^2 = 0$. At $p = 3$ this fails in general.*

Proof. $\mathbb{E}_x \rho^{(e+e')\cdot x} = 1$ iff $e_t + e'_t \equiv 0 \pmod p$ for all t, and 0 otherwise. For $e, e' \in \{0,1,2\}^n$ we have $e_t + e'_t \in \{0, \dots, 4\}$, which for $p \geq 5$ is $\equiv 0$ only if $e_t = e'_t = 0$. So $\mathbb{E}(\Phi_1 - a_0)^2 = \sum_{e,e' \neq 0} a_e a_{e'} \mathbb{E}\rho^{(e+e')\cdot x} = 0$. At $p = 3$, $1 + 2 \equiv 0$: the constant-input characters and the sign-flipped characters are complex conjugates ($\omega^{2x} = \overline{\omega^x}$), and the pseudo-variance is unconstrained. □

Lemma 13 (circular cone bound)**.** *Let Z be a complex random variable on a finite probability space with $\mathbb{E}Z = A$, $\mathbb{E}(Z - A)^2 = 0$ and $Z \in c + K_\alpha$ almost surely, where $K_\alpha = \{|\operatorname{Im} w| \leq \tan\alpha \operatorname{Re} w\}$ and $0 < \alpha < \pi/4$. Then*

$$\operatorname{Re} A - c \;\geq\; \sqrt{\cot^2\alpha - 1}\,\sigma, \qquad \sigma^2 := \tfrac{1}{2}\mathbb{E}|Z - A|^2.$$

Without circularity, for any Z with $Z \in c + K_\alpha$, $(1-t^2)\mathbb{E}|Z-A|^2 \leq 2t^2(\operatorname{Re} A - c)^2 + (1+t^2)\operatorname{Re}\mathbb{E}(Z - A)^2$, $t = \tan\alpha$.

Proof. Put $X = \operatorname{Re} Z - c \geq 0$, $Y = \operatorname{Im} Z$, so $|Y| \leq tX$. Let $E = \mathbb{E}|Z - A|^2$ and $P = \operatorname{Re}\mathbb{E}(Z - A)^2$; then $\operatorname{Var} X = (E + P)/2$ and $\operatorname{Var} Y = (E - P)/2$. Hence $(E - P)/2 \leq \mathbb{E}Y^2 \leq t^2\mathbb{E}X^2 = t^2\big((\operatorname{Re} A - c)^2 + (E + P)/2\big)$, which is the second claim; with $P = 0$ and $E = 2\sigma^2$ it gives $\sigma^2(1 - t^2) \leq t^2(\operatorname{Re} A - c)^2$, the first. □

Theorem 14 (no $p = 3$ mechanism for $p \geq 5$)**.** *Let $p \geq 5$ and θ be single-frequency. With $k_p := \sqrt{\cot^2(\pi/p) - 1}$ and $\sigma^2 := \tfrac{1}{2}\mathbb{E}|\Phi_1 - \mathbb{E}\Phi_1|^2 \;\geq\; \tfrac{1}{2}\sum_{\emptyset \neq T \subseteq [n]} |\beta_{1,T}|^2$,*

$$h(\theta) \;\leq\; \kappa_{\min}(p)\,\big(\operatorname{Re}\overline{\beta_1} - k_p\,\sigma\big), \qquad \gamma(\theta) \;\leq\; \kappa_{\min}(p)\,B_p\,\Lambda_{n,s}(k_p),$$

where $\Lambda_{n,s}(k) := \sup\big\{(\operatorname{Re} A - k\sqrt{E/2})/(\sum_i |\beta_i|^r)^s\big\}$ over finitely many atoms $\beta_i \in \mathbb{C}$, $\xi_i \in \mathbb{T}^n$, with $A = \sum_i \beta_i$, $E = \sum_{\emptyset \neq T} |\sum_i \beta_i \xi_i^T|^2$. $\Lambda_{n,s}$ is non-increasing, $\Lambda_{n,s}(k) \geq \nu^{1-s}$, and $\Lambda_{n,s}(k) = \nu^{1-s}$ for all $k \geq k^*(n,s) := \sup(\operatorname{Re} A - \nu^{1-s}(\sum |\beta_i|^r)^s)/\sqrt{E/2}$. Consequently, if $k_p \geq k^*(n,s)$, no single-frequency network at (p,n,b) beats the IF value of a single feature.

Proof. By Lemma 3 (with $h > 0$, else nothing to prove), $h = \kappa_{\min} \min_x \operatorname{cm}(\Phi_1(x))$, and $\Phi_1 \in c + K_{\pi/p}$ with $c = \min_x \operatorname{cm}$. Lemma 12 and Lemma 13 with $\alpha = \pi/p < \pi/4$ give $c \leq \operatorname{Re}\mathbb{E}\Phi_1 - k_p\sigma$, and Parseval with Lemma 11 gives $2\sigma^2 = \sum_{e \neq 0} |a_e|^2 \geq \sum_{T \neq \emptyset} |\beta_{1,T}|^2$. For the second bound, divide by $\|\theta\|_{2,b}^\nu \geq (C_b^{-1}\sum_i |\beta_{i1}|^r)^s = B_p^{-1}(\sum_i |\beta_{i1}|^r)^s$ and use $\beta_{1,T} = \sum_i \beta_{i1}\xi_i^T$ (paper, Lemma 6) and $\operatorname{Re}\overline{\beta_1} = \operatorname{Re}\sum_i \beta_{i1}$. The IF block gives $\Lambda \geq \nu^{1-s}$. Atoms with $E = 0$ satisfy $\operatorname{Re} A \leq \nu^{1-s}(\sum |\beta_i|^r)^s$ by the paper's Lemma 7, and for atoms with $E > 0$ the inequality $(\operatorname{Re} A - k\sqrt{E/2}) \leq \nu^{1-s}(\sum |\beta_i|^r)^s$ for all $k \geq k^*$ is the definition of k^*. □

Proposition 15 (matrix reduction)**.** *$k^*(n,s) \leq \sqrt{2}\,\kappa_M(\nu,s)$, where*

$$\kappa_M(\nu,s) := \sup\Big\{\frac{1 - \nu^{-s}\|H\|_{S_r}}{\|\operatorname{strictlower}(H)\|_F} :\; H \in \mathbb{C}^{\nu\times\nu},\; H_{jj} = 1\Big\}$$

and $\|H\|_{S_r} = (\sum_k \sigma_k(H)^r)^{1/r}$.

Proof. Rotate the atoms so that $A > 0$ (this does not change E or the cost, and $\operatorname{Re} A \leq |A|$). Let $m_i = (1, \xi_{i1}, \xi_{i1}\xi_{i2}, \dots, \xi_{i1}\cdots\xi_{in})$ and $H = \sum_i \beta_i m_i m_i^*$. Its diagonal is A, and its strictly lower entries are the interval sums $\sum_i \beta_i \xi_i^{(k,j]}$, distinct non-empty patterns, so $\|\operatorname{strictlower} H\|_F^2 \leq E$. As $\|\beta_i m_i m_i^*\|_{S_r} = \nu|\beta_i|$ and $\|X+Y\|_{S_r}^r \leq \|X\|_{S_r}^r + \|Y\|_{S_r}^r$ for $0 < r \leq 1$ (McCarthy, Rotfel'd), $\sum_i |\beta_i|^r \geq \nu^{-r}\|H\|_{S_r}^r$, i.e. $\nu^{1-s}(\sum |\beta_i|^r)^s \geq \nu^{-s}\|H\|_{S_r}$. Hence $A - \nu^{1-s}(\sum |\beta_i|^r)^s \leq A(1 - \nu^{-s}\|H/A\|_{S_r}) \leq \kappa_M \|\operatorname{strictlower} H\|_F \leq \sqrt{2}\kappa_M\sqrt{E/2}$. □

5 Anatomy of the counterexample

For the certified $p = 3$, $n = 11$ network (normalised so that cm is the ratio to the IF value), the field $\Phi = \Phi_1$ satisfies:

clean amplitude $A = \mathbb{E}\Phi$	$2.3960 + 0.0138i$
cone margin $c = \min \mathrm{cm}(\Phi)$	$1.05440 = 0.440\ \mathrm{Re}\, A$
interference energy $\mathbb{E}\lvert\Phi - A\rvert^2$	$258.4\,\lvert A\rvert^2$
pseudo-variance $\mathbb{E}(\Phi - A)^2$	$(-72.2 + 10.0i)\,\lvert A\rvert^2$
Var Im Φ/Var Re Φ	1.776
$\max_x \lvert\Phi(x)\rvert$	$4491\,\lvert A\rvert$
count classes with $\lvert\arg(\Phi - c)\rvert > 59.9°$	21 of 78

The network gives up 56% of its clean amplitude and carries interference two orders of magnitude larger than the feature, but the interference is aligned with the two edges of the 60° cone: on those inputs the correct logit ties (up to the margin) with one neighbouring class and the third class is pushed far down. This is possible only because the fluctuation is *non-circular* (Var Im > Var Re, allowed since $\tan 60° > 1$), and that is possible only because $2 \equiv -1 \pmod 3$ (Lemma 12). In exchange it uses 8 neurons instead of $\nu = 12$, and $(12/8)^{s-1}$-type savings in the concave ℓ_r cost outweigh the lost amplitude.

6 Numerical evidence beyond $p = 3$

All numbers in this section come from local optimisation (L-BFGS with an annealed soft-min of the margins) and are *evidence*, not certificates; scripts are in the supplementary folder.

1. The robust constant $k^*(n, s)$. Lower bounds on the supremum in Theorem 14, at $b = 2$ ($s = \nu/2$), from optimising over atoms (general chains for $n \le 7$, slot-symmetric chains above):

n	2	3	5	7	9	11	13
$k^*(n, \nu/2) \ge$	0.345	0.543	0.764	0.928	1.055	1.073	1.099

p	5	7	11	13
k_p	0.946	1.820	3.256	3.932

The growth flattens. If these lower bounds are close to the supremum, Theorem 14 excludes beating the IF value with single-frequency networks for $p \ge 7$ and all $n \le 13$, and for $p = 5$ and $n \le 7$. For the matrix constant of Proposition 15 we found $\kappa_M(3, 1.5) \approx 0.289$, $\kappa_M(4, 2) \approx 0.476$, $\kappa_M(6, 3) \approx 0.694$.

2. The L^2 bound is far from tight pointwise. At $p = 5$, $n = 9$ the best atom configuration for $(\mathrm{Re}\, A - k_5\sigma)/\nu^{1-s}C^s$ reaches 1.186 ($m = 7$ atoms, $E = 0.79|A|^2$). Realised as a single-frequency network, its cone efficiency is -31, and local optimisation from it (constant input components allowed) never reaches a positive margin. The pointwise cone constraint is much stronger than its second moment.

3. IF behaves as a local maximum. Starting from the IF block plus noise, with the cone widened to any half-angle between 36° and 89°, local optimisation in the full single-feature class (arbitrary input vectors) returns to the IF value (0.9999) at $p = 3$ and $p = 5$, $n = 11$. The $p = 3$ counterexamples live in a different basin, reached only through a structured family of $\approx 0.7\nu$ neurons.

4. Narrowing the cone kills the counterexample. The certified $p = 3$, $n = 11$ network evaluated with a half-angle of 59° instead of 60° has cone efficiency -218 (and -2438 at 50°), because its interference sits on the cone edges.

5. Fewer than ν neurons. The counterexamples save cost by using fewer than ν neurons for one feature. Best normalised margin (ratio to the IF value) over 24 random restarts of slot-symmetric networks with exactly $m = \nu - 1$ neurons:

(p, n)	$(3,3)$	$(3,4)$	$(5,2)$	$(5,3)$	$(5,4)$	$(5,5)$	$(5,6)$	$(5,7)$	$(7,2)$	$(7,3)$
ratio	0.599	0.597	≤ 0	≤ 0	0.278	0.310	≤ 0	≤ 0	≤ 0	≤ 0

At $p = 3$, networks with at most $\nu - 1$ neurons classify at every tested n ($2 \leq n \leq 14$), keep $\approx 60\%$ of the IF value already at $n = 3, 4$, and form the branch that crosses 1 at $n = 11$. At $p = 5$ such networks were found only at $n = 4, 5$, at 28–31% of the IF value; at $p = 7$ none were found for $n \leq 5$ (a continuation run from $n = 5$ to 14 also found none). Random restarts are weak at larger n (at $p = 3$ they too miss the structured branch beyond $n \approx 6$), so this is evidence, not proof.

6. Cross-feature compensation. Conjecture 17 can only fail through features that use each other's slack. We started from the IF network with the programme allocation $y^\star$ (several features), added noise, optionally deleted neurons, and optimised the true margin over all p^n inputs (general, not slot-symmetric networks):

(p, n), features K^*, IF width	from IF + noise	one neuron deleted	two deleted
$(5, 2)$, $K^* = 2$, 6	1.000000	0.837	0.782
$(7, 2)$, $K^* = 3$, 9	1.000000	0.980	–
$(7, 3)$, $K^* = 2$, 8	0.999995	0.764	–

(Ratios to the IF value; best of 3–4 seeds.) In every case the optimiser returns to the IF network, and removing neurons costs margin; $0.782 = \kappa_{\min}(5)/\Gamma_{3/2}(5)$ is the single-feature IF network. Random restarts at $(5, 2)$ with widths 4–8 (30 runs) never exceeded 0.782. No cross-feature gain was found.

7 Corrected statement of Appendix D

Theorem 16 (replaces Conjecture 20)**.** *For all odd primes p, $n \geq 1$ and $2 \leq b \leq \nu$,*

$$\nu^{1-s} B_p \Gamma_s(p) \;\leq\; \pi_{n,b}(p)\, B_p\, \Gamma_s(p) \;\leq\; \gamma_b^* \;\leq\; B_p \Gamma_s(p),$$

with equality in the middle at $p = 3$. The first inequality is strict at $p = 3$, $b = 2$, $n \in \{11, 13, 14\}$ and on the b-intervals of Corollary 7; it is an equality for $b_0 \leq b \leq \nu$ (previous notes, Part II) and at $p = 3$, $n = 2$ for all b (Part I).

Conjecture 17 (price–programme equality)**.** $\gamma_b^* = \pi_{n,b}(p)\, B_p\, \Gamma_s(p)$*: interference changes the price of a feature but not the programme.*

Question 18 (is $p = 3$ the only exception?)**.** Is $\pi_{n,b}(p) = \nu^{1-s}$ for every $p \geq 5$? Theorem 14 gives a positive answer for single-frequency networks whenever $k_p \geq k^*(n, s)$. Open are (a) single-frequency networks with $k_p < k^*(n, s)$ (numerically $p = 5$, $n \geq 9$), and (b) networks whose neurons carry the partner frequencies $2l$, $3l$ of a feature l: these are the only way to make a feature field non-circular at $p \geq 5$, and by the second part of Lemma 13 a large interference needs a large positive pseudo-variance.

Suggested wording for the paper. Section 4 ("How many features"): "Whether a network carrying interference can beat the class value is an open conjecture" → "Interference can beat the class value: at $p = 3$ and $\nu \geq 12$ a single feature can be bought below the interference-free price (Appendix D). This changes the price of a feature, not the allocation: the best networks built from cheaper features still follow the programme (Theorem D.2)." Limitations: "The exact optimum is proved for interference-free networks; for all networks we prove $\pi_{n,b}(p) B_p \Gamma_s(p) \leq \gamma_b^* \leq B_p \Gamma_s(p)$, where $\pi_{n,b}(p)$, the price of a feature, equals the interference-free price ν^{1-s} near $b = \nu$ and exceeds it at $p = 3$, $\nu \geq 12$, $b = 2$."

8 Open problems

1. Conjecture 17. The obstacle is cross-feature compensation: a feature field may leave its own cone where another feature has slack, and at $b = 2$ one neuron can feed several features at no budget cost.

2. Question 18, in particular single-frequency networks at $p = 5$ with $n \geq 9$, where the numerical value of $k^*(n, s)$ exceeds $k_5 = 0.946$, and partner-frequency networks at every $p \geq 5$.

3. The growth of $\pi_{n,2}(3)\nu^{s-1}$ in n (numerically $1.054, 1.18, 1.11, 1.43$ for $n = 11, \ldots, 14$, using the best networks found).